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Dominic Starfish Prime

The Effects of High-Altitude Nuclear Testing

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Declaration

We declare that this project and report is our own work.

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List of Acronyms

D-T Deuterium-Tritium.
DTIC Defense Technical Information Centre.

EMP Electromagnetic Pulse.

GIC Geomagnetically induced current.
GRB Gamma Ray Burst.

HANE High Altitude Nuclear Explosion.
HEMP High-Altitude Electromagnetic Pulse.

NATO North Atlantic Treaty Organisation.

OST Outer Space Treaty.

PTBT Partial Test Ban Treaty.

1 Introduction

The latter half of the 20th century was dominated by the Cold War, characterised by the United States and Soviet Union's scramble for technological and military superiority. The technology most central to the conflict was the nuclear bomb, with its capability for immense destruction demonstrated by the United States at the end of the Second World War[1]. This display of force began a cycle of nuclear proliferation, with both sides seeing its potential as a military weapon and tool for strategic deterrence[2]. In the midst of the Cold War, failure to stop this escalation heightened tensions between the United States and Soviet Union, driving the nuclear arms race to unknown ground and testing bombs with new designs in new areas of the atmosphere[3].

Operation Dominic was a series of atmospheric nuclear tests conducted in 1962 by the United States culminating in Starfish Prime, a test that would fundamentally change the understanding of the Earth's atmosphere and the threat posed to it by nuclear weapons. Starfish Prime was a High-Altitude Nuclear Explosion (HANE) launched from Johnston Island via Thor missile, detonated with a yield of 1.4 megatonnes at an altitude of 400km. Starfish Prime was carried out to investigate the effects of a thermonuclear charge detonated in the ionosphere.

The test location in the upper atmosphere meant that typical effects of nuclear detonations such as a shockwave, mushroom cloud and fireball were not observed, as was predicted[4, 5]. However, the electromagnetic and atmospheric effects largely exceeded their anticipated reach. The detonation resulted in a strong and far-reaching electromagnetic pulse, as well as the formation of a diamagnetic cavity and artificial auroras whose intensity surpassed expectations[4–7].

Despite the precautions taken in choosing the test site, the HANE still damaged infrastructure in space and on the ground, including the break down of satellites and the disruption of the electrical power grid as far from air zero as Hawaii[8, 9]. Starfish Prime drew attention to the lack of understanding amongst nuclear powers on the unpredictability of the weapons they were stockpiling and testing, especially when detonated in non-standard environments.

The years following the Starfish Prime test saw major scientific discoveries as well as crucial treaties signed, protecting mankind and the environment from damage due to nuclear testing. The test also shifted the international perspective on outer space, which is now treated as a fragile environment for the peaceful progress of all people, and protected to that end[10]. The threat due to an EMP is still frequently considered when planning for the security of nations as part of military strategy[11]. Given the scale of its EMP, Starfish Prime is an important data point used to inform military models to this day.

This report examines the Starfish Prime nuclear test from the political environment which preceded its conception, to its detonation and unique physical effects, as well as the widespread consequences that have stretched across the whole globe over nearly a century.

2 Politics and History

To fully understand the significance of Operation Dominic and the Starfish Prime test, it is essential to first consider the political and historical environment in which these experiments were conceived. High-altitude nuclear testing did not arise solely from scientific curiosity, but from the strategic pressures of the Cold War, where technological demonstrations of nuclear

capability served both military and diplomatic purposes.

2.1 Cold War Climate

In one of the final major acts of aggression in the Second World War, the USA used an atomic bomb as a weapon of war for the first time in human history and with it kick-started the Nuclear age. The bombing of Hiroshima and Nagasaki came just weeks after the initial trinity test in Los Alamos, New Mexico, where the first atomic bomb was developed and tested. The event immediately changed global dynamics with the demonstration of a power never seen before [1].

The bombings resulted in the deaths of tens of thousands of Japanese civilians, devastating the carefully selected urban areas. These attacks are the only use of atomic bombs for aggressive purposes and heavily contributed to the Japanese surrender on August 15th 1945, essentially ending World War II.

The implications of atomic bomb usage went far beyond the battlefield. The U.S. Department of State's Office of the Historian describes the concept of atomic diplomacy as the use of nuclear capability to influence postwar negotiations and strategy. Immediately after WWII there were many debates within the US government on how the US's Nuclear monopoly might affect their diplomatic relations with other world powers, particularly the Soviet Union. Some believed that the US's nuclear arsenal would strengthen their negotiation position on a geopolitical level, however, others worried that the possession of atomic weapons would intimidate other powers into developing such weapons of their own [2].

A contemporary letter written by Manhattan Project scientist Luis Alvarez during the Hiroshima mission shows how the atomic bomb immediately transformed international politics. Writing from aboard a B-29, Alvarez explained that a single aircraft had dropped a bomb equivalent to "15,000 tons of high explosive," making large bombing raids obsolete and allowing one plane to destroy an entire city. He warned that nations would now have to "get along together in a friendly fashion, or suffer the consequences of sudden sneak attacks," revealing an early awareness that nuclear weapons had entered diplomacy. His words foreshadow the post-war balance of power and the coming rivalry between the United States and the Soviet Union [12].

In the period immediately following WWII, cooperation between the Soviet Union and USA deteriorated rapidly as the ideological differences between the countries became the central theme in their relationship. The following period was known as the Cold War: the prolonged geopolitical rivalry between the USA and its Western allies and the Communist Soviet Union and their Eastern Bloc Partners. The rivalry shaped geopolitics for half of the 20th century until the dissolution of the Soviet Union in 1991 [13].

Scholar John Lewis Gaddis, in The Cold War: A New History, situates this conflict squarely in the context of the post-World War II environment, where wartime cooperation gave way to distrust. Gaddis notes that by the war's end, disagreements over the future of Europe, the division of Germany, and control over Eastern Europe produced deep and enduring political and ideological divides [14].

According to Gaddis, the Cold War rivalry was more than a series of isolated disputes; it derived from systemic competition between two distinct world views and political orders. The United States attempted to promote democratic capitalism and market economies, whereas the Soviet Union championed communist systems and state control. These differences influenced the countries military strategy and alliances such as the formation of NATO and the Warsaw Pact

[14].

2.2 Nuclear Arms Race

By the mid-1950s the cold war had morphed into a nuclear arms race between the USA and Soviet Union with both countries entering a dangerous phase with the development of thermonuclear (hydrogen) weapons. The USA was the first nation to detonate an H-Bomb in 1952 with the testing revealing power on a scale far greater than the weapons used at the end of WWII. The Soviet Union was not far behind with their own test just three years later in November 1955. These higher-yield weapons represented not only technological milestones but also significant political signals in the Cold War: each successful test was a demonstration of scientific prowess and strategic deterrence to rivals and allies. Large scale tests broadcast capability to the world and formed a central component of superpower efforts to maintain credibility and influence in the evolving bipolar order [15].

The frequency and scale of the tests increased throughout the 1950s as both sides aimed to improve their designs at a faster pace than the other. The yield sizes of the bombs reached the megaton range and testing was conducted across many different sites, particularly in the Pacific and at the Nevada Test Site, as part of an intense period of weapons development. This period of increased testing led to growing international concern forcing political efforts to curb the nuclear testing [15].

During the testing era of the mid to late 1950s, policymakers and diplomats began to formulate a plan to cease testing on a global scale. In October 1958, the USA announced a moratorium on nuclear tests, the offer was conditional on other nuclear powers following suit. The moratorium was intended as the first step in broader global arms control [16].

The moratorium was driven by a number of factors. Across the world there was huge public anxiety over nuclear war. The effects of nuclear war and its associated fallout became well know by the late 1950s. Scientific research into the effects of nuclear weapons on the atmosphere and pressure from peace movements encouraged leaders to pursue measures that would reduce open-air explosions and testing. The Geneva test-ban talks sought to formalise these pauses into enforceable frameworks. While the moratorium was put in place, it was voluntary and unenforceable relying heavily on the good faith of nuclear powers rather than the strength of international law. The moratorium did, however, bring about a temporary pause in open air testing. Despite this, technical programmes contained behind the scenes further developing thermonuclear weapons [17].

The moratorium unsurprisingly collapsed in August 1961 when the Soviet Union withdrew from the agreement and resumed testing. Shortly after, the USA also pulled out the agreement with their own series of testing beginning in September of the same year. The Soviet Union's resumption of testing included a series of extremely high yield bombs, culminating in the detonation of the Tsar Bomba on the 30th of October 1961. The blast had an estimated yield of 58 megatonnes, making it the largest bomb ever tested and still remains to this day the highest yield bomb ever tested. The timing of the Tsar Bomba test, coming just weeks after the collapse of the moratorium, was widely interpreted as an explicit show of strength by Nikita Khrushchev's government [18].

The tested escalated tensions between the USA and the Soviet Union, instead of representing a return to routine weapons development, the collapse of the moratorium and the Soviet Union's hammer blow of Tsar Bomba served as a stark symbol of the underlying mistrust and rivalry that defined the nuclear arms race [19].

2.3 Motivation for Operation Dominic

The Dominic series of tests in 1962 was not the first resumption of US nuclear testing post moratorium, in fact, the USA had resumed testing underground soon after the collapse of the agreement. The series known as Operation Nougat began on 15 September 1961 at the Nevada Test Site, consisting primarily of underground detonations used to continue weapons development with minimal atmospheric fallout. These tests occurred in close at a similar time to the Soviet Union's own resumption of nuclear detonations, following its abandonment of the voluntary test pause [20].

Operation Dominic, conducted in 1962, was the major component of the renewed United States' atmospheric and Pacific testing programme. The series of tests was scheduled following the collapse of the treaty as a way to signal strength and maintain momentum in weapons development. Dominic was a large and rapid series of experiments investigating new designs and effects of weapons [3].

The decision to launch Operation Dominic was shaped by several intertwined political objectives. Operation Dominic was executed against the backdrop of a failed series of global arms control efforts. Officials in the Kennedy administration viewed continued atmospheric testing as a way to demonstrate U.S. resolve and technical capability in the face of Soviet advances. Contemporary accounts describe the programme's planning as "rapid and ambitious," emphasising that one of its driving forces was the political need to respond in kind after the Soviet Union ended the moratorium and resumed its own tests. The tests were also intended to reassure allies of American military power and deter adversaries with a visible demonstration of strength [21].

State Department documents from the period also refer to Dominic as fitting into the larger sequence of post-moratorium testing, framing it as complementary to the resumed U.S. tests that began in late 1961. This context shows that the series was not a spontaneous scientific endeavour, but a calculated demonstration of capability designed to project strength during a period of intense superpower rivalry [3].

A central concern of US government officials was maintaining the credibility of their nuclear arsenal. Continuous testing was seen as essential to ensure the deterrent value of nuclear weapons. Reports from the era indicate that Dominic included a significant number of weapons design tests intended to validate advanced systems and address potential gaps revealed by years without atmospheric detonations. The testing served as political and technical goals as it signalled to allies and rivals that the United States remained at the forefront of nuclear science. Thus, Operation Dominic can be understood as part of a broader strategy aimed at sustaining U.S. credibility in the nuclear balance, which was central to Cold War efforts.

The decision to undertake Operation Dominic was also influenced by domestic political pressure. Within the US military, leaders pushed for a robust testing agenda arguing that national security relied on vigorous continuation of nuclear development especially after the Soviet Union resumed their own testing when they broke the moratorium. While direct declassified state documents specifically linking internal U.S. political pressures to Dominic are limited in open-source form, the historiographical record suggests that Pentagon advocacy and institutional momentum behind testing contributed to the timing and scale of the series [22].

Contemporary international political forums, such as the United Nations First Committee debates on disarmament, also reflected growing global scrutiny and domestic criticisms of re-

sumed testing.

While Operation Dominic was shaped by political pressures the consequences of Starfish Prime ultimately depended on the weapons that were tested. Exploring the effects observed during the test is vital to gain a full understanding of its impact.

3 Nuclear Weapons

The Dominic Starfish Prime high-altitude nuclear test involved the detonation of a thermonuclear weapon, a high-yield bomb that utilised both nuclear fission and fusion. The specific architecture and yield of thermonuclear weapons are central to understanding the extreme electromagnetic and radiative effects observed during Starfish Prime.

To understand the design principles and motivation for developing thermonuclear bombs, fission weapons must first be explored. The earliest designs of nuclear weapons were developed purely around the process of nuclear fission. These weapons, commonly known as 'Atom Bombs', derive their explosive power from a rapidly diverging neutron-induced chain reaction. There are two distinct fission types, spontaneous and induced fission reactions. In the context of nuclear weapons, the latter are much more important. In an induced fission process, a large, unstable nucleus splits into two or more nuclei upon absorbing a neutron, simultaneously releasing vast amounts of energy, neutrons, and radiation. One of these possible fission processes is shown in Figure 1 for a Uranium 235 nucleus. The general equation for this fission reaction is below.

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow \mathrm{A} + 2.45\,^{1}_{0}\mathrm{n} + 202.5\ \mathrm{MeV} \qquad (1)$$

An average of 2.45 neutrons are emitted in this

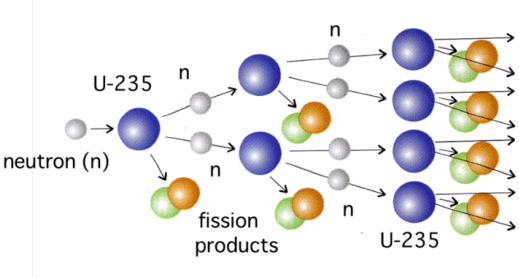


Figure 1: **Neutron induced fission of uranium-235**. Emitted neutrons lead to a chain reaction fission process. Adapted from [23].

decay [24]. The term A represents the possible fission fragments produced, most frequently two daughter nuclei are created with a proton-to-mass number ratio similar to the parent nucleus which is known as the unchanged charge distribution rule [25]. The average total energy release by slow-neutron induced fission of a uranium-235 nucleus is 202.5 MeV, or 83.14 TJ/kg [26]. Fission reactions like the one above release vast amounts of energy compared to chemical reactions. Take the oxidation of hydrogen, used in hydrogen fuel cells, which for the same mass releases a millionth of the energy produced by uranium-235 fission [27]. The origin of this energy release can be understood in the change in nuclear binding energy between the reactants and products, with the binding energy defined as,

$$B(N,Z) = (Zm_H + Nm_n - M(N,Z))c^2 \qquad (2)$$

where Z is the number of protons, m_H is the mass of a hydrogen nucleus, N is the number of neutrons, m_n is the mass of a neutron, c is the speed of light, and M(N,Z) is the neutral atomic mass.

The highly energetic neutrons produced during a fission reaction can interact with surrounding fissile material to further induce fission. Where nuclear weapons and reactors differ is the control of the rate of neutron production. Nuclear reactors carefully moderate the balance of neutron production and loss to ensure a steady

energy output [28]. In contrast, a fission bomb is designed to drive the neutron population into rapid, exponential growth within a fraction of a microsecond.

3.1 Criticality

The evolution of the neutron population can be written as

$$N_{n+1} = kN_n \tag{3}$$

where k is the effective neutron multiplication factor and N_n is the number of neutrons in the nth generation of the chain. It's often quantified as the ratio of the rate of neutron production to the rate of neutron loss in a system [29]. A nuclear system's k value determines its behaviour,

$$\begin{cases} k < 1 & \text{subcritical:} \\ & \text{The neutron population exponentially decreases.} \\ k = 1 & \text{critical:} \\ & \text{The neutron population maintains a steady value.} \\ k > 1 & \text{supercritical:} \\ & \text{The neutron population exponentially increases.} \end{cases}$$

In nuclear reactors, the k value has very small oscillations around 1 [32]; conversely in a fission bomb, the k value can significantly exceed values of 1 before decreasing as the explosion occurs.

The critical mass is defined in nuclear engineering as the minimum mass of fissile material required to produce a sustained nuclear chain reaction [29], that is have k = 1. It depends on the fissile materials fission cross-section, density, shape, enrichment, temperature, and surroundings. The shape of fissile material that maximises criticality is a perfect homogeneous sphere. Keeping with the earlier definition for the effective neutron multiplication factor, to maximise k, the ratio of production to loss of neutrons must be maximised. Neutrons are produced throughout the bulk of the material, so the production rate scales with volume, while neutron losses occur primarily at the boundary and therefore scale with surface area. A spherical geometry maximises the volume-to-surface-area ratio, minimising neutron leakage and hence maximising the effective neutron multiplication factor [29].

The higher the density of fissile material, the lower the critical mass. Fissionable atoms are closer together when in a densely packaged arrangement, increasing the probability of neutrons colliding with them. The density of fissile material can be varied by the pressure exerted upon it, a principle that forms the basis of the

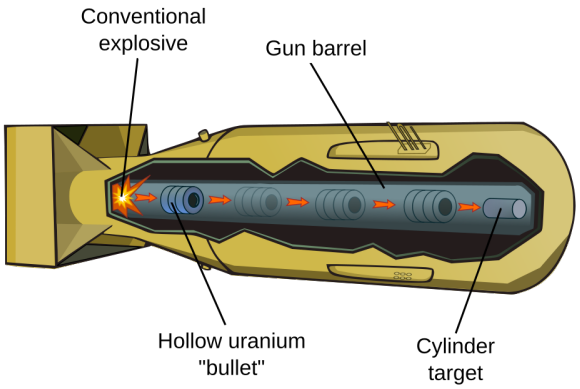


Figure 2(a): **Gun-type fission bomb**. The conventional explosive propels the sub-critical 'bullet' of uranium towards the target, making the assembly supercritical and igniting the bomb [30].

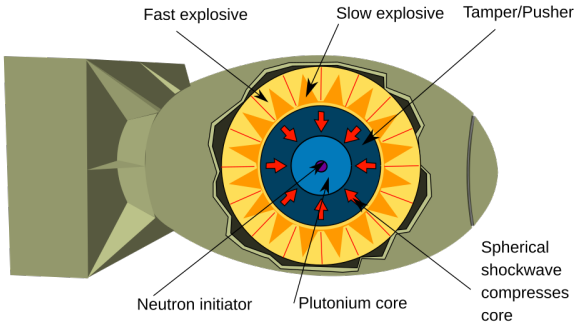


Figure 2(b): **Implosion fission bomb**. The explosive lens's explode synchronously and create and a large implosion pressure that compresses the core, making the plutonium core supercritical [31].

implosion design, explored further in the next section.

Fission bombs employ neutron reflectors to surround the core of fissile material to reduce the required critical mass, and increase the probability of subsequent fission in an already critical mass [33]. These shells, often made of materials like beryllium or heavy metals, elastically backscatter escaping neutrons into the core, reducing the likelihood of neutrons escaping.

3.2 Types of Fission Weapons

In a nuclear weapon, the mass of the fissile material must be kept subcritical until detonation is desired. In the past, there have been numerous fatal accidents of subcritical fission cores being accidentally made critical, most notably the demon core at Los Alamos [34]. For use in nuclear weapons, plutonium-239 is favoured over uranium-235. The primary benefit is the significantly smaller spherical critical mass required than that of uranium-235, with its untampered critical mass of 10.0 kg compared to uranium-235's 46.7 kg [35]. Thus, the quantity of plutonium-239 required for a nuclear weapon is much less than of uranium-235. Furthermore, the process of enriching naturally occurring uranium (increasing the ratio of U-235 to U-238) is both more costly and longer than the production of Pu-239 from U-238. Finally, the average number of neutrons produced by the fission of a single atom of Pu-239 is 2.91 [36], greater than the 2.45 produced in the fission of U-235.

The first type of fission bomb to be produced was the gun-type weapon, where a piece of subcritical material is shot into another sub-critical piece, creating a supercritical mass [32]. Typically, the target mass of highly enriched uranium was a cylinder, while the part that was fired was a cylindrical shell that would encompass the target. The supercritical configuration is achieved by assembling a sufficient quantity of uranium such that the dimensions of the mass span many neutron mean free paths, making it highly probable that most fission neutrons induce further fission events before escaping.

The gun-type bomb faced a multitude of issues and as such is obsolete in the modern-day. Firstly, it must use highly enriched uranium because plutonium isotopes have high spontaneous fission rates [37], which when used in this design would lead to pre-detonation. They also suffer from low efficiency as they need a very large mass of fissile material and a relatively low percentage of the material actually fissions due to the slow timescale of the assembly.

These shortcomings gave rise to the Implosion nuclear weapon. The design of this weapon has a sub-critical sphere of fissile material, known as the pit, surrounded by a tamper and pusher shell. Surrounding these shells are uniformly spaced explosive lenses. A device known as an initiator is added to the core of the pit, which produces free neutrons when compressed [38]. The explosive lenses simultaneously detonate, creating an implosion shock wave that compresses the pit and increases its density to supercriticality [32].

These bombs also employ a tamper, a dense shell of material that surrounds the fissile core. This shell's inertia contains the thermal expansion of the fissioning material, keeping it supercritical for longer. The tamper thus lengthens the reaction, and makes for a more energetic explosion [39]. The tamper is frequently constructed from depleted uranium which can fission via high energy neutrons produced by the principal explosion. This can increase the yield considerably.

3.3 Thermonuclear Weapons

The need for more powerful and efficient bombs led to the development of thermonuclear bombs, colloquially know as a hydrogen bomb, the second generation nuclear weapon. They employ both fission and nuclear fusion to release vast amounts of energy, with the first hydrogen bomb's yields exceeding 20 times that of the most powerful fission bombs [40].

The essential nuclear reaction for a thermonuclear weapon is the Deuterium-Tritium (D-T) fusion reaction, with the heat and pressure of fission, a deuterium atom fuses with a tritium atom.

$$ {}^{2}_{1}\mathrm{D} + {}^{3}_{1}\mathrm{T} \rightarrow {}^{5}_{2}\mathrm{He}^{*} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV} \quad (4) $$

While the energy output is roughly a tenth of the fission reaction in Equation 1, the reactants are only a 15th as big. Consequently, the energy output per unit mass is approximately 5 times as large. The neutrons produced in this reaction carry 80% of the energy produced, and can induce fission in uranium or plutonium atoms [40].

Another important nuclear reaction is the production of tritium as shown below. Lithium atoms under neutron bombardment split to give a tritium atom.

$$ {}^{6}_{3}\mathrm{Li} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{3}_{1}\mathrm{T} + 5\ \mathrm{MeV} \quad (5) $$

This reaction is used to generate the tritium used for the fusion with deuteride in Equation 4.

3.4 Teller-Ulam Configuration

The principal design of a thermonuclear weapon is the Teller-Ulam configuration, a bomb designed with a primary and secondary stage. Though unconfirmed, it is likely that the W49 warhead used in the Starfish Prime nuclear test had this configuration. Much information on the design of these weapons is still classified, a possible design of the bomb is shown in Figure 3a. The primary stage is a modification of a small implosion fission bomb as described in subsection 3.2. At the centre of the pit there is a small amount of fusion fuel, included for enhancing the efficiency of the primary. Upon heating and compression, the fusion fuel releases excess neutrons which induce further fission. The secondary stage consists of a column fuel, frequently lithium deuteride, encased by multiple layers [43]. Surrounding the column of fuel is a

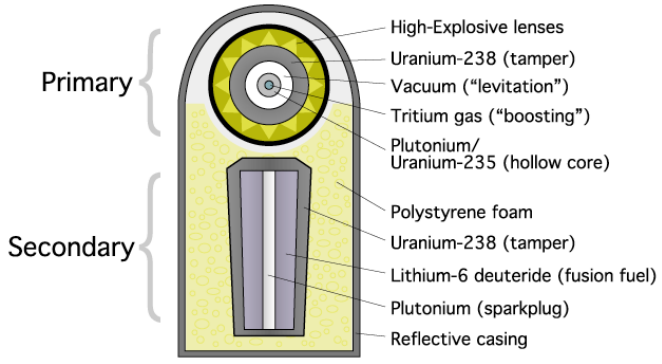


Figure 3(a): **Possible Teller-Ulam configuration**. The primary stage consists of a small implosion fission bomb. The energy produced by the primary induces the secondary, igniting a D-T fusion reaction. Labels indicate components [41].

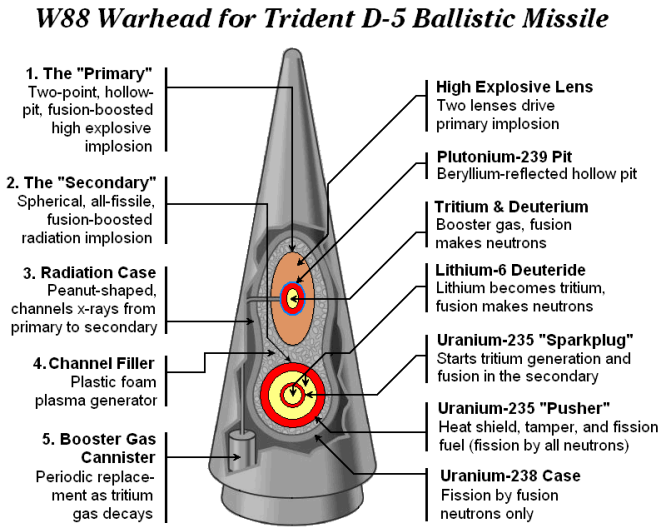


Figure 3(b): **W88 American thermonuclear warhead**.This design uses the Teller-Ulam configuration. Labels of components with short descriptions. Adapted from [42].

pusher-tamper, a heavy layer of uranium-238 or lead which serves a similar function as described in subsection 3.2. Within the fuel column is a spark plug, a column of subcritical fissile material (typically plutonium-239 or uranium-235) which helps ignite the secondary. Surrounding the two stages and forming the outmost casing of the bomb is a radiation case, made of lead or uranium-238. This design is shown in figure Figure 3a.

After the primary's implosion fission bomb has run to completion, its temperature is several million degrees and radiates gamma and X-rays [44]. The principal function of the radiation case is to contain the X-rays emitted by the primary and redirect them toward the secondary. While some X-rays are elastically backscattered, the dominant mechanism involves absorption by the radiation case, leading to substantial heating before the energy is re-emitted toward the secondary. These X-rays exert considerable radiation pressure on the secondary, however, they are not the primary source of compression [44]. The outer layer of the pusher-tamper is heated to extreme temperatures by the X-rays, such that it expands violently and ablates away [43]. Considering total momentum conservation, the recoil from the surface layer ejection creates a pressure an order of magnitude stronger than radiation pressure. This whole process is known as radiation implosion. This compression forces the uranium spark plug supercritical and it undergoes fission, further heating the fusion fuel. The secondaries fuel then begins the D-T fusion reaction and a fireball forms. The neutrons produced by the D-T fusion reaction and by fission of the U-238 spark plug induce further fission in the fissile material that makes up the pusher-tamper and neutron reflector, contributing significantly to the bombs yield [43]. It is this multi-stage fission–fusion–fission process, and the enormous neutron and gamma flux it produces, that underpins the extreme electromagnetic phenomena

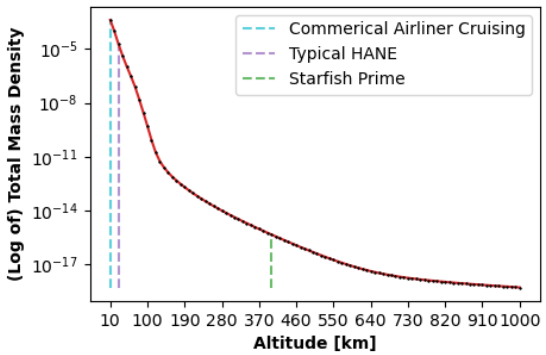


Figure 4: **Log plot of air density in the atmosphere with increasing height**. Data measured on June 9th 1962 at 23:00 UTC (H – 11 hours Honolulu time [5]). Raw data acquired from NASA's NRLMSIS 2.1 empirical global reference atmospheric model [45], and plotted using matplotlib with Python. Dotted lines represent 10km, 30km and 400km from left to right. At 30km the atmosphere is ~ 10 orders of magnitude thinner than at 10km and at 400km the atmosphere is ~ 11 orders of magnitude thinner than at 10km.

observed during the Starfish Prime high-altitude test.

4 High Altitude Testing

Nuclear detonations on or near the ground infamously produce bright fireballs, mushroom clouds and strong shockwaves. These phenomena all require a dense atmosphere to propagate; a fireball needs oxygen, a shockwave needs an air pressure gradient, and a mushroom cloud requires chaotic temperature gradients combined with heavy radioisotope contamination [46]. Earth's atmosphere thins rapidly as altitude increases, illustrated in Figure 4; so what happens when a thermonuclear charge detonates in a sparse region of the atmosphere?

The X-rays, beta emission, gamma rays and neutrons released at air zero (the point of det-

onation in the air) are dispersed uniformly and travel great distances unburdened in the upper atmosphere. The asymmetry of the atmosphere relative to air zero leads to a strong angular dependence in the absorption mechanisms. Radiation on a path towards Earth will be absorbed by the air in the exponentially thickening atmosphere, as seen in Figure 11, whereas radiation pointed towards the stars will ultimately be absorbed by satellites or caught in Earth's magnetic field [4]. On the ground a minimal fireball masked by a bright flash followed by an auroral display will be observed [5]. The consequences of releasing these radiation types into the atmosphere is discussed below.

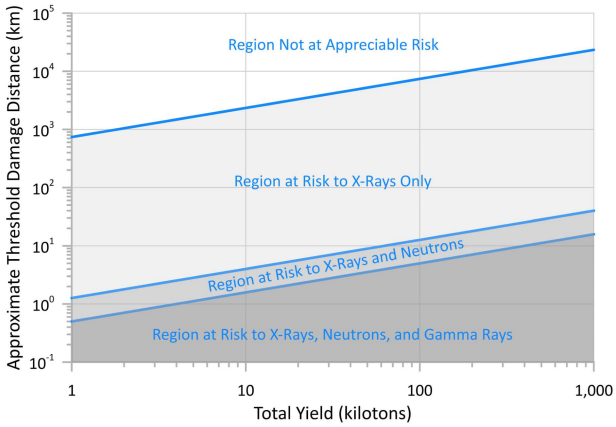


Figure 5: **Illustration of satellite susceptibility to various prompt effects**. The energy released in the detonation directly affects the energy and quantity of radiation emitted so the distance is taken as a function of detonation yield. The model is an estimate formed using empirical data, no real data points are present. Air zero and the burst point refer to the same location. For reference, the yield of Starfish Prime was 1400 kilotons [47]. Figure from [4].

4.1 Prompt Radiation

Prompt effects are those which are a direct consequence of the radiation released by the bomb and typically do not persist for long after detonation. The prompt effects at these high altitudes differ significantly from those observed after ground detonations. The typical mushroom cloud and shockwave are impossible without a dense body of air which circulates the air currents or mediates a shockwave. The fireball exists only intermittently and is restrained to a much smaller volume than those closer to the ground [4, 5].

A vast majority of the energy released in a two stage thermonuclear charge is released in the form of X-rays (around 70% [4]) so these effects encompass a lot of the damage potential. Human infrastructure at these high altitudes is limited to public-use and military satellites (and the ISS) so damage to infrastructure is relatively contained. X-rays incident on satellite bodies cause excessive heating which eventually results in a shock wave capable of fracturing components aboard [4]. Further damage is incurred by the build up of ionised atoms on the surface of satellites which induces a large spike of current across on-board electronics [4]. The remaining 30% of the total yield is split up amongst gamma-ray, beta and neutron emission. Large neutron and gamma ray fluxes displace atoms in the outer coatings of satellites which deteriorates functionality of onboard equipment and significantly shortens the projected lifespan of the satellite [4]. Gamma radiation is the source of the EMP effect which will be discussed in further detail in section 5 as well as the associated effects on infrastructure and the atmosphere in section 6. The effect of beta radiation is largely the Argus effect, which is a persistent effect that is discussed in Figure 4.3 and section 7. The susceptibility of commercial satellites to the various forms of damage incurred by exposure to prompt beta, gamma and neutron radiation are illustrated in Figure 5.

4.2 Atmospheric Composition

Before discussing the persistent effects, it is necessary to outline a few key components of the atmosphere at high altitudes. The aspects necessary to discuss within the scope of this report are the ionosphere and the natural radiation belts. The ionosphere is a layer of ions trapped in the upper atmosphere at altitudes between ~ 60km and ~ 20000km. The ionosphere is split up into regions which are formed via distinct interactions with solar radiation. The exact mechanisms for the formation of these layers lies outside the scope of this project and are thus omitted. An illustration of the different regions and their altitudes is depicted in Figure 6. Due to the strong correlation with solar activity, the ionosphere is observed to share an oscillation period with both the Earth (annual) and the Sun's solar cycle (~ 11 years). An empirical model of the dynamics inside the ionosphere's layers was developed for use in industry, military and educational purposes, called the International Reference Ionosphere [48].

The Earth's magnetic field traps two radiation belts in orbit, called the Van Allen Belts, which are separated by a vast region of nearly unoccupied space [49]. The inner radiation belt overlaps with the uppermost layers of the Ionosphere but the regions are strictly distinct; the ionosphere is the plasma layer whereas the radiation belts are the trapped energetic particles. The upper radiation belt lies much higher than the ionosphere and consists of particles which originate primarily from the sun [49]. As with the ionosphere, several empirical models are available for computing the dynamics of the Van Allen belts, of which the most up to date is the AP9/AE9 model developed by NASA [50].

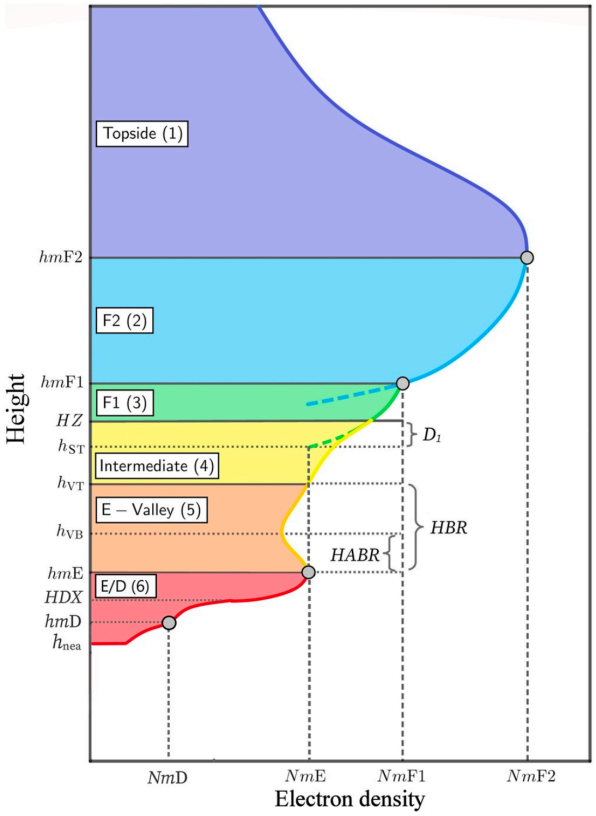


Figure 6: **Plot of electron density in the ionosphere against altitude**. The various regions are labelled on the y-axis. The splitting of the F regions occurs only during sunlight hours [48].

4.3 Persistent Effects

The persistent effects predominantly affect the atmosphere and are thus explained in greater detail in section 7. The most significant persistent effect is known as the Argus Effect [4]. Depending on the altitude of the detonation this effect is a result of various mechanisms local to the detonation - for example, for a detonation at an altitude of 30km the X-ray and beta emissions heat up the air around air zero causing it to rise up to the natural radiation belts [4]. This injection of charged particles into the natural radiation belts is the Argus effect.

In the case of Starfish Prime, the Argus effect was detected via elevated levels of radio

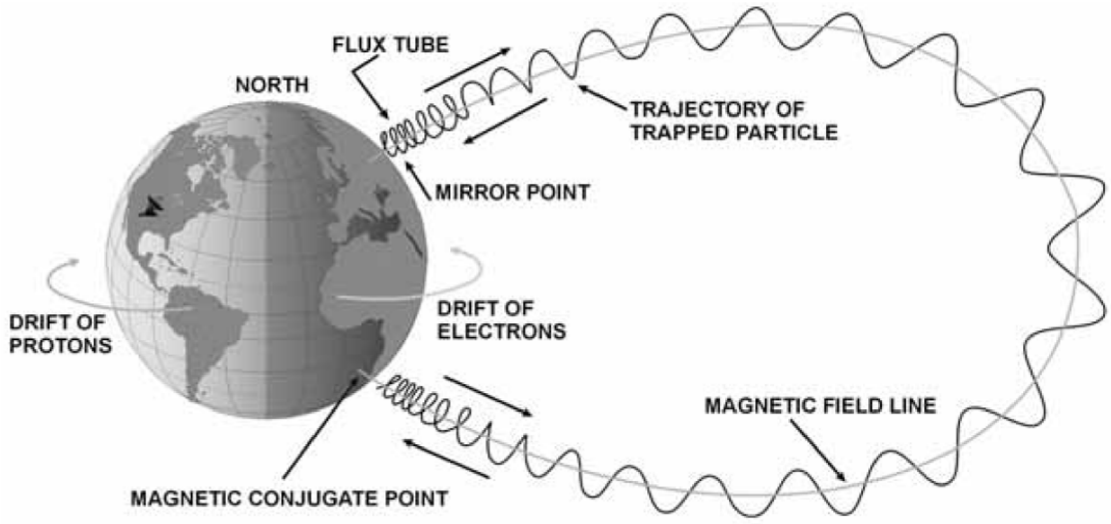


Figure 7: **Dynamics of trapped ions in Earth's magnetic field**. The toroidal motion about the field lines results in synchrotron emissions in the radio frequency range. Figure from [51].

frequency synchrotron emissions [5], which occur due to the toroidal path of charged particles as they travel along Earth's magnetic field lines. This effect is depicted in Figure 7. Observable deviations from the norm were present for up to 5 years after the detonation [7]. The ionosphere is widely used in radio communication systems across the world so the disruption of this atmospheric layer is a key point of concern. Ions created during detonation alter the absorption levels of different frequencies of electromagnetic radiation by the various layers of the ionosphere [52]. For example, after one of the Operation Argus tests, a 5 hour blackout was observed because the D region of the ionosphere had a significant spike in absorption of radio frequency radiation triggered by the high altitude detonation [53].

4.4 Dominic Starfish Prime

Starfish Prime was one of five nuclear tests conducted during Operation Fishbowl [54]. At the time of the operation the exact outcome of a high altitude high yield nuclear detonation was uncertain. However, prior testing with high altitude low yield detonations during Operation Argus [55], fuelled expectations for a long lasting aurora and ionospheric disturbance [5]. The main objectives of the test series were to better understand the electromagnetic effects concerning communications and surveillance equipment; to investigate the aspects of the detonation which could directly or indirectly harm humans, and to evaluate the effectiveness of current nuclear detection and diagnostic systems.

4.5 Location of the Test

Air zero was chosen due to its proximity to the unpopulated Johnston Island and the success of prior testing in the region [56]. At the time there was fear that the bright flash after detonation could cause temporary or permanent blindness to civilians who directly observed the blast [56], so air zero was positioned to the South West of Johnston Island, away from Hawaii. Fig-

ure 20 shows the nearby islands positions relative to Johnston Island. Despite this precaution, public infrastructure on Hawaii was affected by Starfish Prime's strong EMP [8]. The exact coordinates of air zero were 16°28′ N 169°38′ W, at an altitude of 399km [5, 8, 54]. The payload was delivered via a Thor missile from a launch pad situated on Johnston Island [5].

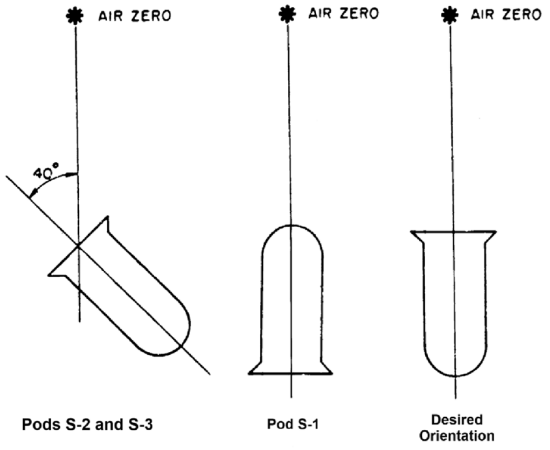


Figure 8: **Orientation of pods S-1, S-2 and S-3 relative to their desired orientations for optimal x-ray measurements**. The backplate which is normal to air zero has the instrumentation flush on that surface. Pod S-1 measured no data at all due to complete misalignment. Pod S-2 and S-3 deviated from the normal by 43° and 40° respectively. Pod S-3 was ~ 9*km* further from air zero than intended so suffered from both shadowing (~ 50% of the desired flux) and also the ~ $\frac{1}{r^2}$ spherical distribution of X-rays. Pod S-2 suffered only the shadowing due to misalignment so contained the best data of the three. Published yield calculations were not performed with this data [5]. Modified figure from [5].

4.6 On-board Data Collection

The Thor missile which carried the Starfish Prime payload carried with it three telemetry pods, each weighing approximately 600kg, fixed to fins near the bottom of the rocket [5]. These pods contained various instruments but were primarily designed to measure the effects of prompt X-ray radiation at a close proximity to the detonation (within ~ 14km). The pods had no ability to control their position so projectile dynamics were relied on for correct orientation. Figure 8 displays the shortcomings of this approach. The near total failure of the pods to capture useful X-ray data was compensated by measurements performed by third parties using a mix of rocket-borne, airborne, ground-based, seaborne and satellite instruments. For example, the total yield of the bomb was calculated using one of the only intact total X-ray yield datasets available, which was measured by a rocket-borne calorimeter operated by Sandia Corporation [4]. These data were used by scientists to develop a clearer picture of the complex processes which occur when a thermonuclear charge detonates at high altitudes.

5 EMP Theory

An electromagnetic pulse (EMP) is one of the main consequences of high altitude nuclear detonations where effects such as a shockwave, mushroom cloud and fireball are not present for significant periods of time. The primary driver for its generation are the prompt X-rays, gamma rays and neutrons released during an exoatmospheric nuclear explosion as described in section 4. In general, an EMP is a short burst of electromagnetic energy, also referred to as a transient electromagnetic disturbance. Its causes can be both natural, such as lightning strikes, or man-made, such as nuclear explosions [57].

The high-altitude EMP (HEMP) is vital to consider in the case of Starfish Prime. It consists of three electric field components - early-time (E1), intermediate-time (E2), and late-time (E3) [58]. While the HEMP does not pose a direct threat to humans, human infrastructure,

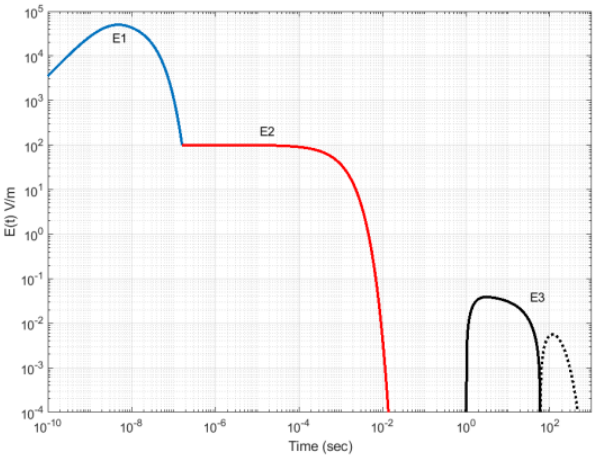


Figure 9: **Variation of the electric field magnitude of a generic high altitude pulse with time elapsed since detonation**. The wave form is separated into E1 (blue), E2 (red) and E3 (black) EMP signals, the dashed line represents a reversed sign on y-axis. Figure from [61].

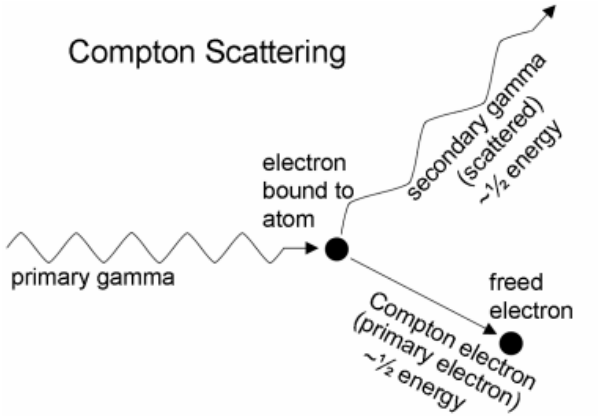


Figure 10: **Diagram of the Compton Scattering process**. Gamma rays from the detonation scatter off electrons bound to atoms in the air. Secondary Gamma rays (on average) are scattered in the same direction so as to form a forward facing gamma absorption layer. Figure from [62].

such as electrical systems responsible for daily life, is particularly at risk [59, 60].

Consequently, a detonation near an urban area would lead to a disruption of the electric power grid, telecommunications and other services vital to normal function and recovery from a nuclear attack. This section explores the HEMP formation mechanisms as well as the range and magnitude of the EMP before introducing the effects that EMPs have on various aspects of human infrastructure which are then explored in more detail in section 6.

5.1 Early-time EMP (E1)

The early-time HEMP is the first stage of the electromagnetic burst generated by a nuclear weapon that explodes above the atmosphere. It is a short lasting, narrow pulse as seen in Figure 9 that lasts up to approximately 1μs after the burst, arising from the prompt gamma rays released in the initial explosion [61]. These gamma rays travel down in an expanding shell of several meters, eventually reaching altitudes of 20 to 40km where the atmosphere is much denser. At this height, collisions of gamma rays with air atoms become significantly more frequent, causing the gammas to deposit some of their energy here [58, 62, 63]. This is known as the gamma absorption layer which is illustrated by the yellow layer in Figure 11.

When a gamma ray strikes an electron of an air atom within the gamma absorption layer, it transfers about half of its energy to the electron, liberating it from the atom. This produces Compton electrons and scattered gamma rays with about half the original energy via the Compton scattering process which can be seen in Figure 10. The Compton electrons are scattered parallel to direction of gamma emission from the burst and continue to travel down towards the Earth. As they do, they experience the Earth's magnetic field which curves their path, as seen in Figure 12. Motion of the charged electron in a magnetic field generates a radiated electromagnetic (EM) field, which travels outwards at the speed of light [62]. At each point along the burst-observer path, increasing amounts of the

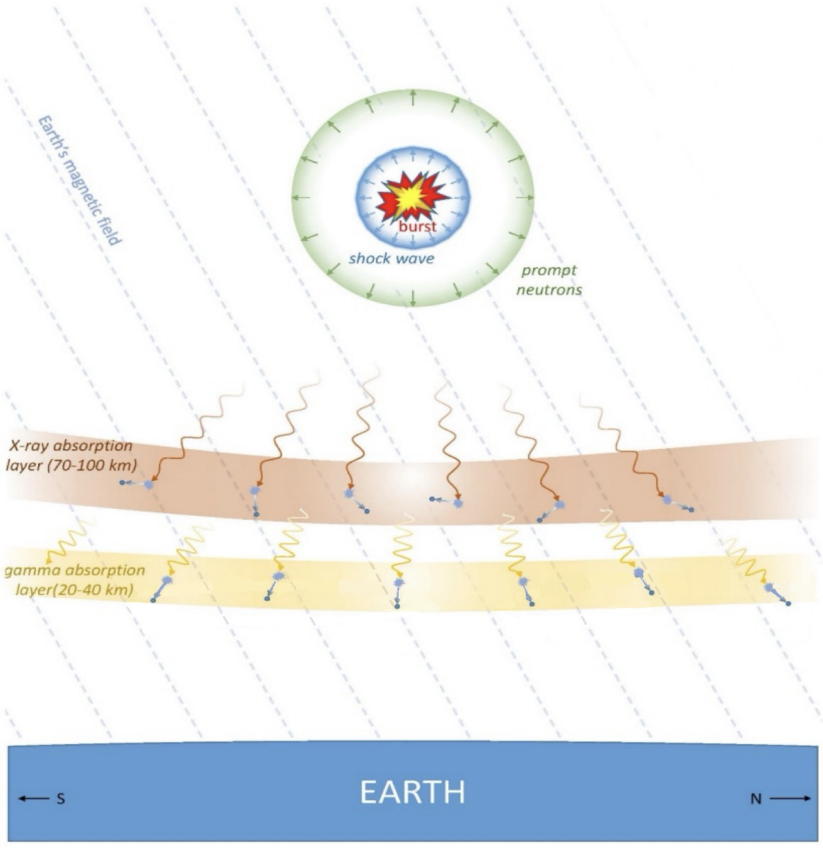


Figure 11: **Gamma and X-ray absorption layers of an exoatmospheric nuclear burst**. The gamma absorption layer (yellow) is also known as the source region and is relevant to the generation of the early-time EMP. The X-ray absorption layer (orange) plays a significant role in the behaviour of the late-time EMP. Figure adapted from [63].

EM field contributions are generated and add up to a net stronger field. Therefore, the strength of the EMP is increased when the distance from explosion to observer increases as more field contributions take part in the formation of the EMP. As the gamma rays travel downwards to regions of higher atmospheric density, they are eventually attenuated and the field no longer increases in magnitude. From this point onwards, the field travels down to earth at a constant magnitude.

The geographic region that will be exposed to the E1 pulse is determined by the line of sight distance between the earth and burst location. Therefore, the altitude of the detonation is directly related to the size of the region that will be affected by E1 [4, 61]. Figure 13 shows an example geographical area affected by an E1 pulse in the northern hemisphere, where the relative strength of the pulse experienced on Earth is indicated using colour. The strength of the E1 is determined by the relative orientation between the geomagnetic field lines and the direction of gamma ray propagation. The null point, which is a local minimum in the electromagnetic disturbance, occurs geographically north of ground zero, indicated in green in Figure 13, when the ray and field lines are parallel. Conversely, the maximum field is experienced in a concentrated area geographically south of ground zero, where the field lines and gamma direction are perpendicular. At this point, the deflection of the electron path is maximised which generates the

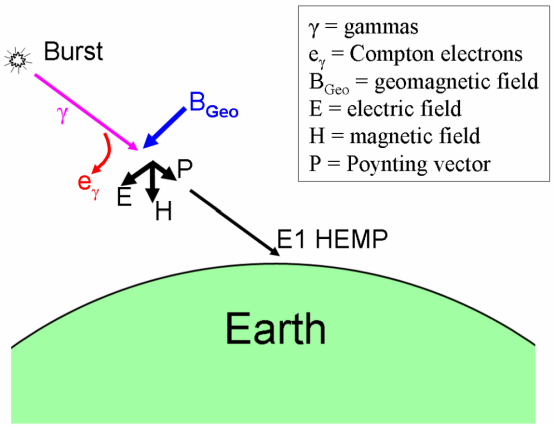


Figure 12: **Formation mechanism of general E1 HEMP**. The gammas from nuclear burst are shown in pink. Via the Compton Scattering process, they liberate electrons whose path becomes curved due to the Earth's magnetic field (red). Their movement through the geomagnetic field generates an electromagnetic field that radiates down as the early-time E1 HEMP (black). Figure from [62].

largest magnitude of E1 EMP [62].

Moreover, the strength is strongly correlated to the angle of entry of rays into the gamma absorption layer. A steeper entry angle, closer to ground zero, experiences a larger peak field. This is because a shorter atmospheric path length is travelled and a more compact region of Compton electron production is sampled, leading to a lower degree of temporal spreading of generated Compton currents, making the E1 peak in Figure 9 higher and narrower [62].Therefore the strength of the E1 pulse depends on the balance of several different factors.

5.2 Intermediate-time EMP (E2)

The next stage of a HEMP is the intermediate-time pulse which follows immediately from E1 and lasts from 1μs to 1s after the detonation [58]. It consists of two stages - E2A and E2B. E2A arises from the gammas scattered during Compton scattering as part of E1 production. E2B pulse is generated when high energy neutrons from the explosion scatter inelastically from air molecules to produce further gamma rays [62].

The characteristics of E2 are comparable to that of a lightning strike in terms of duration and frequency waveform but it is much more geographically widespread, covering an area similar to E1 [60]. Unlike lightning, E2 itself does not physically strike conductors, but propagates through the air and induces transient voltages when it couples to conductors [61]. Generally, E2 only becomes a serious concern if adequate lightning protection systems are not in place or have been severely damaged which can occur due to the initial E1 pulse. As a result, the direct effects of E2 have seldom been studied in depth [60].

5.3 Late-time EMP (E3)

The final stage of the HEMP is the late-time pulse that begins approximately one second after the burst when E1 and E2 play no further role in the HEMP. This component was most significant for Starfish Prime as E3 maximises at a height of burst of 130 to 500km [64]. The E3 EMP is born from the motion of plasma, ionised weapon debris, and ionised atmosphere as a result of the burst. It consists of two phases - blast wave (E3A) and heave wave (E3B).

The E3A initial blast wave is generated in the first 10s after the explosion. During this time, about a quarter of the blast energy from the nuclear device is transferred to the kinetic energy of weapon debris [65]. This ionises the debris from the weapon and the surrounding atmosphere to create a highly conductive ball of plasma [63]. The high energy of this material causes it to expand rapidly from the burst creating a 'diamagnetic cavity', a bubble of plasma

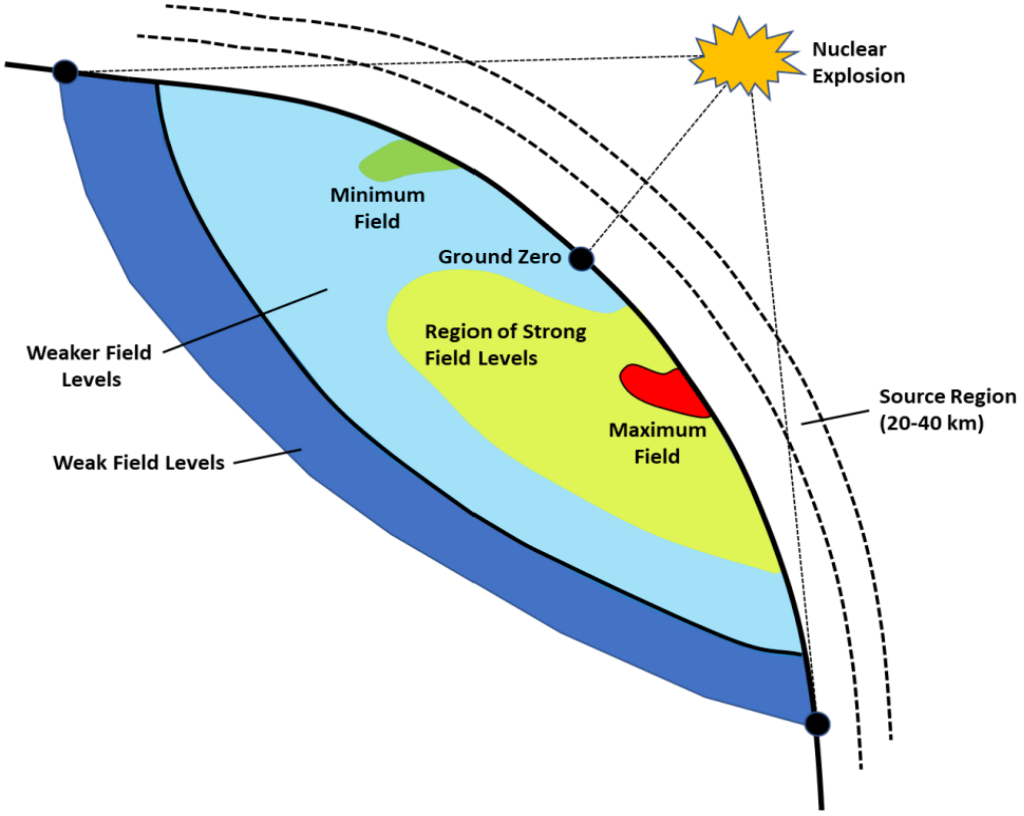


Figure 13: **'Smile' diagram showcasing the area affected by the E1 EMP**. The total geographical region of exposure is defined by line of sight distance from location of explosion. Figure from [61].

embedded in the magnetic field of the Earth in the upper atmosphere. This is shown in Figure 14. The plasma ball is a conductor that is expanding in a magnetic field; thus, internal currents are generated to maintain a constant magnetic flux penetrating its surface. These internal currents create a magnetic field within the bubble that distorts the geomagnetic field lines in order to counteract the changing magnetic flux. This deformation induces electrical fields at the surface of the Earth which are the physical manifestations of the E3A burst pulse [63].

The size and shape of the plasma bubble are controlled by the weapon yield and altitude. A greater yield clearly leads to a larger and more energetic plasma bubble and thus more significant deformations to geomagnetic field lines. The altitude affects the rate of bubble expansion and its shape via atmospheric and magnetic effects [63]. As the diamagnetic cavity grows, the surrounding magnetic field becomes increasingly distorted and strengthened. Further expansion therefore requires more of the debris' kinetic energy to be converted into magnetic potential energy, which progressively slows the growth of the cavity.

Through this mechanism, the diamagnetic cavity develops an asymmetric shape. The cavity requires significantly less energy to expand parallel to the magnetic field lines than perpendicular to them, since perpendicular expansion demands a greater degree of magnetic distortion. The

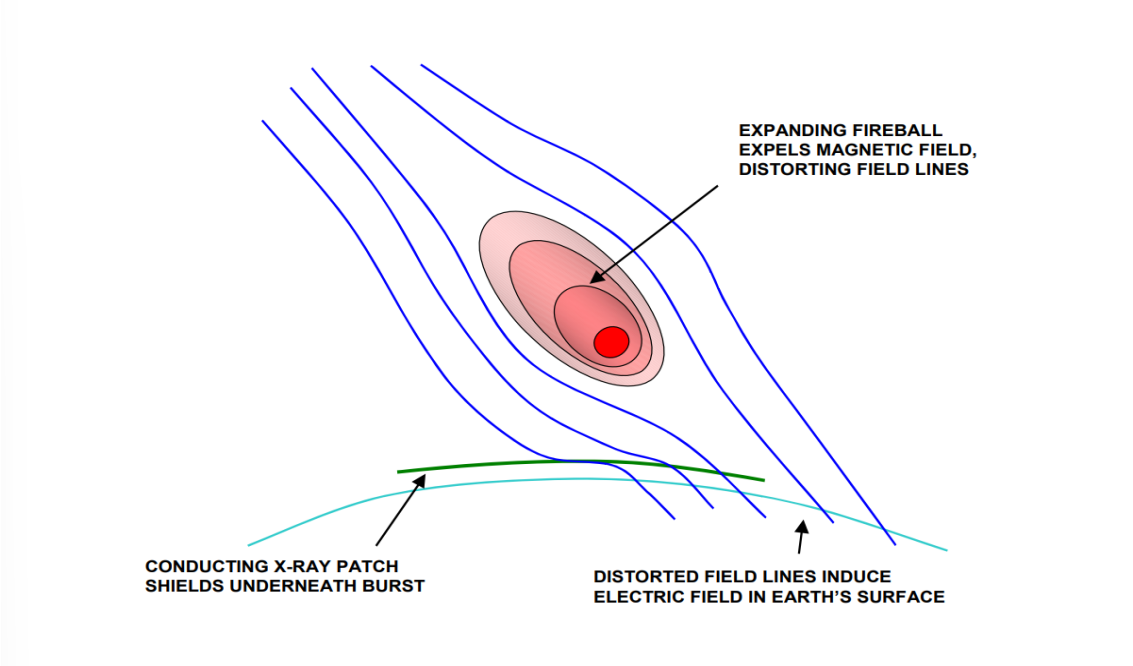


Figure 14: **Diamagnetic cavity created during the formation process of the E3A blast wave**. The geomagnetic field lines (dark blue) around it are significantly distorted and the cavity itself is elongated and aligned with the Earth's magnetic field. The light blue line represents the surface of the Earth. The dark green line represents the X-ray absorption region. Figure from [64].

asymmetry this creates is then exaggerated further by the height dependence of atmospheric density. Downward expansion is progressively more difficult, due to the increasing air density, while upward expansion becomes slightly easier. This causes the cavity to favour upward expansion. Together, these effects produce a highly elongated cavity, aligned with the Earth's magnetic field, as shown in Figure 14.

The X-ray absorption layer plays a significant role in shielding the Earth from E3 EMP. This layer is highlighted in orange in Figure 11 and lies at an altitude of 70-100km [63]. A nuclear burst produces X-rays which are emitted while plasma ball is still hot via Blackbody radiation. These X-rays interact with air atoms via the photoelectric effect which ionises them. The process involves the annihilation of the X-ray photon where all its energy is transferred to one of the atom's electrons so that it can be liberated from the atom. In contrast to Compton scattering, the released photoelectrons do not have a fixed direction so there is a low coherence of electronic motion in the X-ray absorption layer. This causes the layer to heat up and become extremely conductive, an effect which lasts for several seconds after the burst [63]. The absorption layer has the effect of pinning the geomagnetic field lines, shielding the area immediately below the burst for an electromagnetic disturbance. As a result, the largest effect of the E3A is experienced beyond the area covered by the conducting X-ray patch, which is shown in Figure 17.

The second component of the E3 EMP and final component of the total HEMP is the E3B heave pulse. On the heave timescale, from 10s to several minutes [64], the plasma ball has finished expanding and begun cooling such that the X-ray ionised layer decays and no longer shields the Earth. The buoyant air pushes the plasma

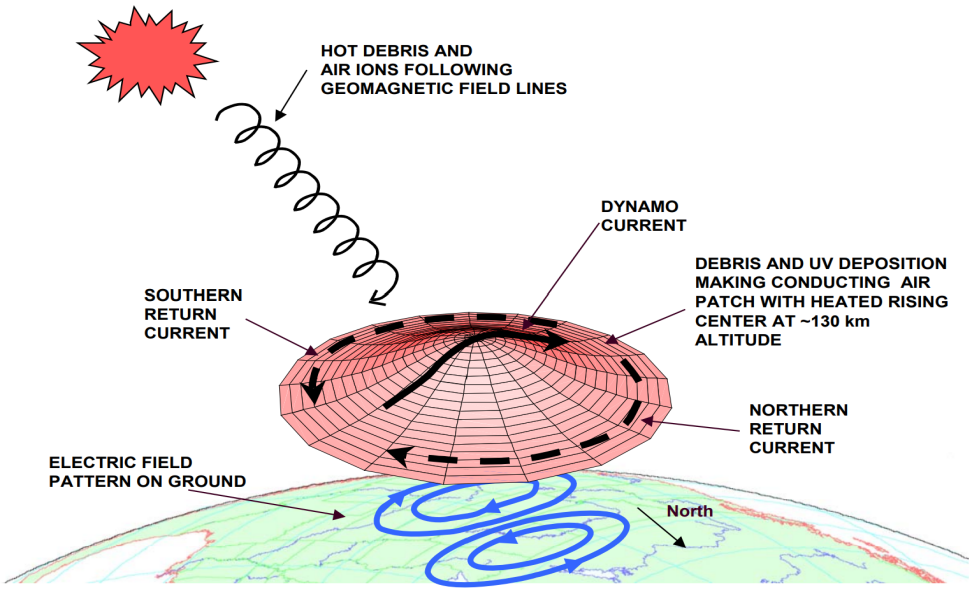


Figure 15: **E3B formation mechanism showing the rise of a heated ion patch**. The ion patch is indicated by the red 'hat' shape at the centre. Below, blue lines indicate the electric field pattern on the ground. Figure from [64].

ball higher in the atmosphere. While the Earth's magnetic field returns to normal, a new conductive patch of ions builds up from the ionised debris and atmosphere ions at an altitude of 130km [63, 64]. Ultraviolet radiation emitted from plasma ball is incident on the ion patch causing heating effects within it. The patch expands and again internal currents are generated which deform the field lines below the patch of ions.

Immediately below the burst, the centre of the patch is most buoyant and begins to rise fastest resulting in the 'hat' shape, shown in red in Figure 15. Induced by the Dynamo effect, the currents in this region lead to rapidly changing magnetic deflections on the surface of the Earth causing large electric fields. The effects of changing fields on the surface are shown in Figure 16, where the Dynamo current manifests as a double bulls-eye pattern. The null field regions lie inside the bulls-eyes as there is a large cancellation between Dynamo and return currents [63]. However, the field is not zero here since the magnetic field behaves dynamically as the bulls-eyes move outwards with the rise of the ion patch. In the case of Starfish Prime, the E3A/E3B data was recorded inside of the bulls eyes and is therefore not useful [63].

It is important to also note that the E3 pulse is stronger closer to Earth's magnetic poles. This is a consequence of the increased geomagnetic field line density at the poles. Here, an expanding plasma ball and ion patch will cross more field lines inducing greater internal currents generation and thus increased E3 strength of the E3 EMP pulse. Eventually, the ion patch cools completely and all magnetic fluctuations seize. At this point, no more EMP effects are experienced on the surface of Earth as all three stages, E1, E2 and E3 are now complete [63].

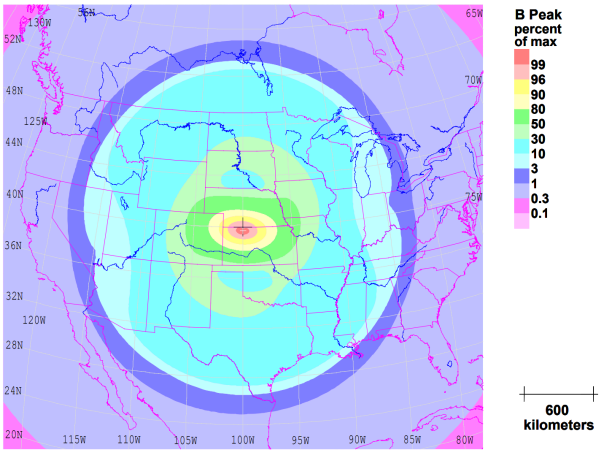


Figure 16: **Spacial distribution of magnetic field fluctuations associated with the E3B (heave) wave for a burst at 130km above continental United States**. The detonation location is indicated by the black cross. Two bulls-eye regions can be seen to the magnetic north and south of the sub-burst point [64].

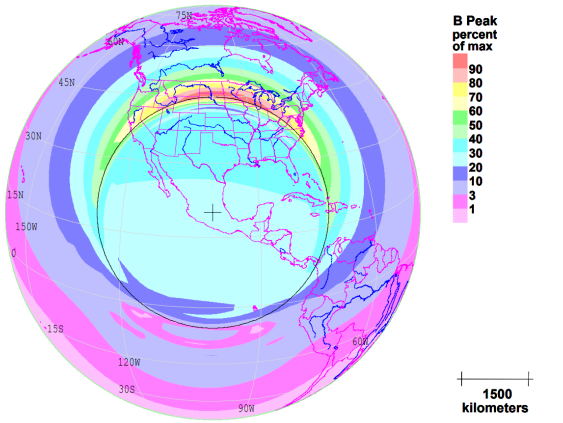


Figure 17: **Spacial distribution of magnetic field fluctuations associated with the E3A (blast) wave for a burst at 400km above Mexico**. Detonation location is indicated by the black cross. The X-ray absorption region is indicated by the thin black circle. Figure from [64].

6 EMP Effects

Starfish Prime was highly significant in the understanding of electromagnetic pulses. The models generated for Starfish Prime drastically underestimated the reach of the EMP, exposing gaps in the understanding of high-altitude nuclear explosions. The inconsistencies between theory and experiment were due to the lack of understanding of the mechanisms that govern the formation of EMPs at high altitude and the effect they have on electronic systems on earth [6].

6.1 General Effects of an EMP

The three previously mentioned stages of an electromagnetic pulse, E1, E2, and E3, all have different effects due to their different properties. The early-time pulse has a large amplitude carrying a content of hundreds of mega-Hertz [62]. It poses two types of threat to electrical systems: conductive and radiated. The first of these considers conductive objects, such as overhead lines or power cables, which are able to absorb the radio frequency energy supplied by the pulse. This causes abrupt changes in voltage and current, known as transients. These cause thermal degradation of electrical component insulation and damage by short circuiting of electrical systems [66]. A radiated threat is when the EM field couples to the equipment to generate voltage and current transients at the level of the device circuit board, causing direct damage to it.

E1 is the highest frequency and fastest component of the EMP, on the order of nanoseconds, which is significantly faster than the lightning timescale [67]. Therefore, not only does the insulation break down, causing short circuiting, but the power surge protection in place for lightning cannot respond quickly enough to shield against the pulse [68]. This makes smaller, more sensitive equipment more susceptible to damage as they cannot withstand large surges in current, so components overheat and get damaged.

The E2 part of the EMP is slightly longer than the E1 section and produces a similar power surge to that created by lightning, so equipment is often already protected against it where protection against power surges from lightning is in place. However, this can be ineffective as the E1 component typically damages these protective systems [60].

The E3 component of the EMP was particularly significant in the Starfish Prime test due to the high altitude of the detonation, which maximised its interaction with the Earth's magnetic field. It is the longest part of the EMP and is created due to the bomb temporarily distorting the magnetic field of the Earth from the rapid expansion of ionised debris. Instead of directly affecting small electronic devices, E3 couples into long conductors such as power lines, inducing low-frequency currents of up to hundreds of thousands of amperes [60].

Systems, such as power and telecommunication lines, span many kilometres, therefore, they are very effective at collecting and conducting E3-induced currents. These are known as geomagnetically induced currents (GICs) which are DC-like. This makes power grids, which are designed to carry AC, particularly vulnerable to damage from a high-altitude nuclear bomb [66]. When GICs enter the windings in power system transformers, the core of the transformer becomes magnetically saturated, reaching its maximum magnetisation. This results in overheating, causing the capacitors to be overloaded, potentially causing grid lockdown or fires [60]. Figure 18 shows a transformer that was damaged by a geomagnetic storm in 1989 by a similar but less intense GIC as induced by an E3. The large coils in the transformers were burned and melted. This type of damage can cause a grid collapse, as it did in 1989 in Quebec, which has significant secondary effects on the distribution of food, water and fuel, as well as issues with communication networks and emergency services [68].

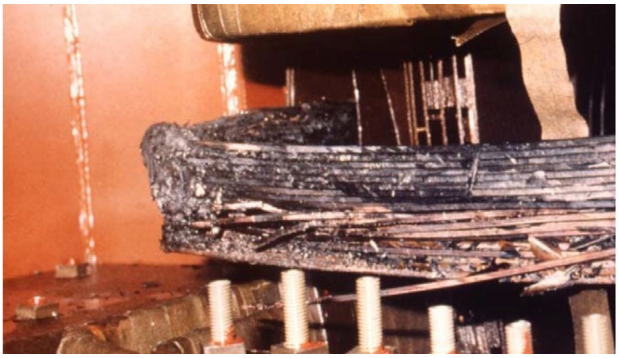


Figure 18: **Damage done to a transformer by a geomagnetic storm in 1989**. This damage was caused by a current of similar magnitudes to that observed by an E3 EMP creating intense localised heating, melting the windings [69].

Generally, the main effects of an electromagnetic pulse occur through conductive structures, such as cables, antennas and transmission lines which couple to the EMP energy causing induced current surges. Any metal object can interact with the pulse but it is the induced current and voltage surges that are responsible for the damage caused to electrical equipment, rather than the EMP itself.

6.2 Effects on Hawaii

As previously mentioned, the extent of the reach of the EMP was underestimated, making the effects in Hawaii, approximately 1450 km away, largely unexpected [6]. The high altitude of the detonation caused the E3 component of the pulse to have the most significant impact on Hawaii. Figure 19 shows the flash of the explosion as see in Honolulu through the cloud cover.

The effects in Hawaii were not just visual. The EMP caused the simultaneous failure of 300 streetlamps on the island of Oahu, as fuses were burned out due to a current surge [67]. Burglar alarms across Honolulu were also triggered



Figure 19: **Flash of the Starfish Prime nuclear explosion seen in the sky in Honolulu**. Image taken approximately 1,450km away from the ground zero [70].

and telephone links between Kauai and other Hawaiian Islands were shut down. Radio stations also went off air temporarily. These disturbances highlighted the susceptibility of communication systems that rely on long conductors. The effects were due to the EMP coupling to the equipment connection lines rather than the devices themselves, minimising permanent damage [67]. The map in Figure 20 shows the location of each effect relative to the detonation point at Johnston Island.

The EMP over Hawaii was relatively weak because the detonation location over Johnston Atoll is close to the equator. The strength of the magnetic field at the explosion site has a significant effect on the strength of the EMP. A high-altitude detonation over the equator results in a lower strength EMP than if the detonation was closer to the magnetic poles of the Earth [68].

Additionally, the damage done to Hawaii was relatively low and largely not permanent due to the types of electronics used in Hawaii in the 1960s. For example, most telephone systems did not shut down as they relied on the use of vacuum tubes instead of semiconductor devices like transistors [6]. Vacuum-tube equipment is approximately 10 million times more resistant to an EMP than solid state circuitry [6]. Solid state electronics are particularly vulnerable as semiconductor components can be burned out by small amounts of EMP energy, only requiring 10^{-7}J to cause damage [67]. Today, modern electronics are highly dependent on semiconductor devices, making the threat of EMPs more prevalent. This suggests that while the EMP effects can be geographically widespread, the severity of damage is strongly dependent on infrastructure and geomagnetic conditions.

6.3 Soviet Union High Altitude Testing

While the effects observed in Hawaii were limited, similar high-altitude tests conducted by the Soviet Union demonstrated that EMP damage could be far more severe when carried out closer to the Earth's magnetic poles. During the Cold War, the Soviet Union's high-altitude nuclear tests were carried out as part of the K

Project nuclear test series. The most significant EMP damage was caused by a detonation at 290km on the 22nd of October 1962, shortly after the Starfish Prime test. The weapon used was much smaller (0.3 megatonnes as opposed to Starfish Primes 1.4 megatonnes [71]) but the damage from the EMP was much greater as it was detonated in a location where the Earth's magnetic field strength was higher and the test was carried out over a populated area rather than the Pacific Ocean, as done in Starfish Prime. The damage to the city of Jezkazgan in Kazakhstan included the blowing of all the fuses in a 570-kilometre telephone line as well as causing a power plant to catch fire in Karaganda, leaving permanent damage. 1000km of buried power cables were also destroyed [71].

Due to the secrecy surrounding Soviet nuclear testing and the subsequent dissolution of the Soviet Union, test data and results are minimal, limiting Project K's impact on the research into EMPs. However, it does demonstrate how destructive an EMP can be when the bomb is detonated over populated land, in a location with a stronger magnetic field.

A year after the US and the Soviet Union high altitude tests, the Partial Test Ban Treaty was passed so Starfish Prime and the K project tests are examples of the very few case studies that we have that demonstrate the effects of an EMP.

6.4 Modern Day Effects of the EMP

Starfish Prime was a key real-world case study for the effects of an EMP, and despite the immediate effects in 1962 being limited, it highlighted the potential threat of an EMP attack. To safeguard against an EMP attack, by 1971, the US military was spending more than 250 million dollars a year on EMP testing to protect electronic systems [6]. However, since Starfish Prime, technology has developed greatly and become far more reliant on solid state electronics which are significantly more sensitive to current surges than the previously used vacuum tube equipment of the 1960s [6]. This implies that an EMP attack today would create much more significant damage.

Modern infrastructure is also now greatly interconnected so damage to one sector could create a ripple effect into many others. For example, power grids, satellite systems, GPS navigation, financial networks, and communication networks all depend on functioning electronic operations [68]. Considering the vast scale of the geographic area an EMP could affect, the potential damage from such an event could be extensive. To ensure minimal damage, aircraft and satellites are routinely checked for EMP resilience. Damage to satellites directly from an EMP is limited but can cause short term electronic disruption. Satellites are instead far more vulnerable to damage from the radiation belt created in a high-altitude detonation which will be discussed in more detail in the next section.

Since Starfish Prime, ways in which EMP threats are mitigated include hardening infrastructure, improving surge protection and transformer resilience, and developing emergency responses to prolonged power outages. Starfish Prime was a critical event in the understanding of EMPs and acted as the catalyst for research into protection against them.

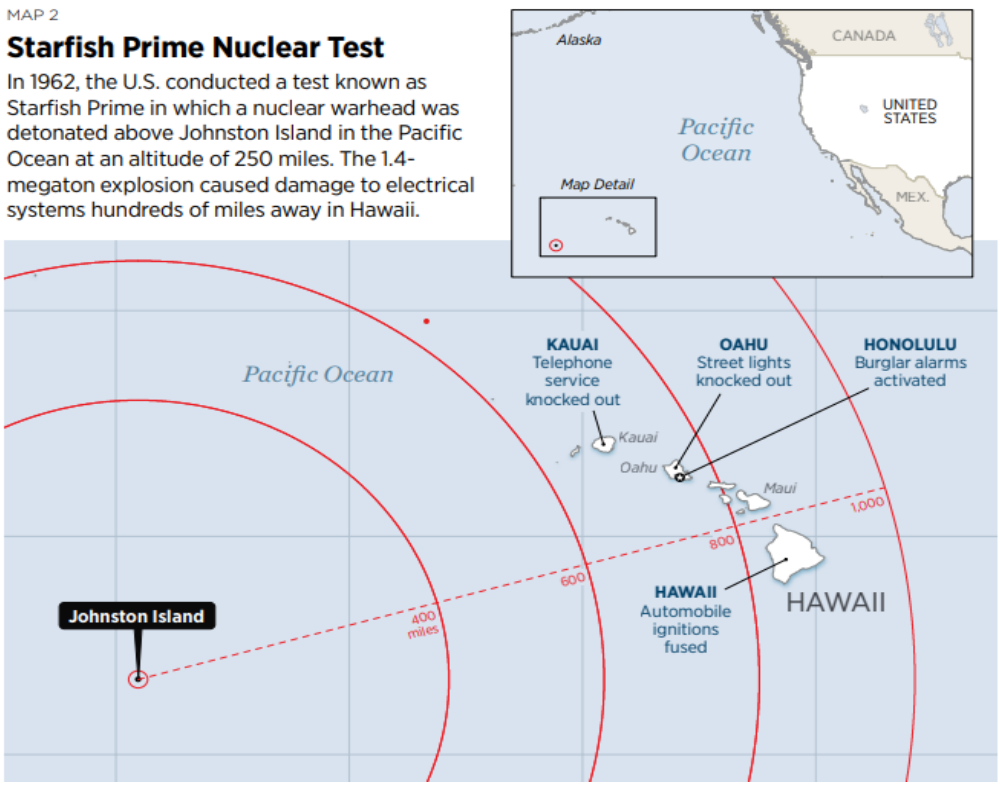


Figure 20: **Map showing the detonation location of Starfish Prime**. Nearby islands, including Johnston Island, are shown relative to the locations in Hawaii which were affected by the EMP [72].

7 Effects on the Atmosphere

In addition to the initial EMP, the Starfish Prime test caused long-range disturbances to the Earth's magnetic field and ionosphere. These disturbances lasted long after the explosion and produced effects such as the disruption of radio communication, changes to the structure of the Van Allen radiation belts, and artificial aurora [73].

7.1 The Diamagnetic Cavity

As discussed previously, the Starfish Prime test created a 'diamagnetic cavity' in the ionosphere, which is illustrated in Figure 14. The Earth's magnetic field was excluded from within this cavity, and heavily distorted outside of it.

The high energy of the Starfish Prime test meant that the cavity expanded very quickly. Data taken on the night showed that the debris expanded outwards at speeds over 1000km/s [74]. This let the cavity reach it's full size within 1.2s, measuring 1840 km along the magnetic field lines and 680 km vertically across [65].

Inside, the expanded diamagnetic cavity was filled with a low-density but hot and highly ionised plasma. This plasma was a mixture of weapon debris, ionised atmospheric particles, and free electrons. The free electrons originated from Compton-scattered atmospheric gas and the beta decay of fission fragments and neutrons released during the nuclear explosion [75]. The electron density within the cavity was significantly higher than that of the surrounding

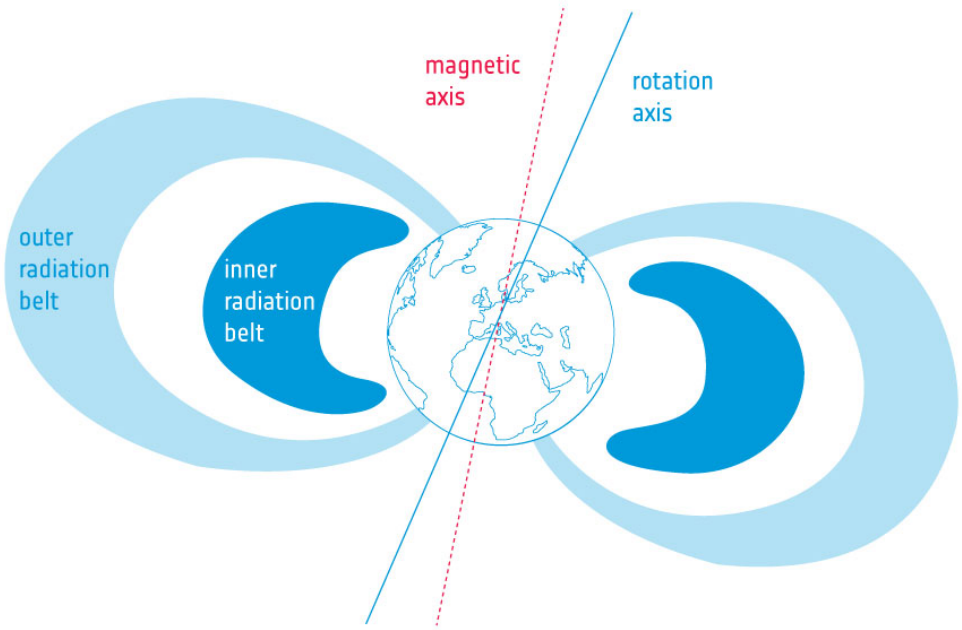


Figure 21: **Schematic diagram of the Van Allen radiation belts**. The inner belt (dark blue) and outer belt (light blue) are separated by the slot region (white). The natural radiation belts consist primarily of trapped energetic protons and electrons, in the inner and outer belts respectively (from ESA [76]).

ionosphere, which lead to a 'nuclear blackout', where radio-based communication was disrupted [77]. This occurred because free electrons would refract, reflect or absorb any radio waves propagating through the plasma [67]. Because the Starfish diamagnetic cavity was extremely large, the vast majority of radio waves passing through it were attenuated or lost, preventing radio communication.

Shortly after the cavity reached its maximum size, it began to collapse, as the outwardly driven plasma pressure was no longer energetic enough to expel the highly distorted magnetic field lines. The collapse took approximately 16.4 seconds to complete, after which, the Earth's magnetic field lines were no longer excluded from the region. During this process, instabilities at the boundaries of the cavity caused charged electrons and ions to be expelled along the geomagnetic field lines [74]. As a result, many energetic particles became trapped within the magnetosphere, forming an artificial radiation belt. Meanwhile, other electrons followed the geomagnetic field lines into the upper atmosphere, producing auroral displays that were visible far from the test site. Both of these phenomena are discussed in more detail in the following sections.

7.2 Artificial Radiation Belt

As described previously in section 4, Van Allen radiation belts are stable toroids of high energy particles trapped in the Earth's magnetic field [78]. There are two main belts: the 'inner belt', which is dominated by high energy pro-

tons, and the 'outer belt' which is dominated by high energy electrons. The inner belt is located between 1.1 and 2 earth radii, while the outer belt extends between 3 and 6.5 earth radii [74, 79]. These two zones are separated by the 'slot region', which is, by comparison, empty of particles. This structure is shown in Figure 21.

The particle concentration within the Van Allen radiation belts can vary very quickly with time. The population is dependent on the balance between 'source processes', which inject particles into the belt, and the 'loss mechanisms', which remove them. Examples of source processes are solar wind and cosmic rays, while 'loss mechanisms' would include particle precipitation into Earth's atmosphere and escape into interplanetary space. The slot region between belts is relatively empty because the loss mechanisms there are much stronger than any source of charged particles.

As briefly discussed earlier, the Starfish Prime test, and the subsequent collapse of it's diamagnetic cavity, brought a huge number of charged particles into the magnetosphere. A large number of electrons were injected into the inner radiation belt and slot region, but those in the latter were quickly lost to space. The loss rate of electrons as a function of distance from Earth is shown in Figure 22. This figure shows that the electron population was particularly stable at 1.7 Earth radii, but in fact the electrons were able to survive anywhere in the range of 1.3 to 1.8 Earth radii. In this stable range, the artificially injected electron population was detectable for at least 5 years [7, 74]. Above and below this range, however, the population decayed over the course of months, as there were more effective loss mechanisms.

This artificial radiation belt increased the electron intensity across the inner radiation belt by several orders of magnitude [80]. This occurred over a much shorter timescale than the lifetime of the electrons in the belt. This meant

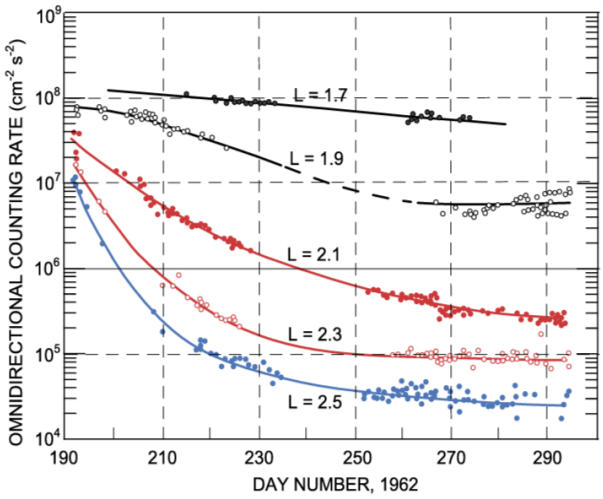


Figure 22: **Count rate of electrons over time after Starfish as a function of distance.**. The figure uses L to denote different distances, which is the distance between the Earth's centre and where the field lines cross the magnetic equatorial plane. The figure shows that electrons are most stable at L=1.7 (solid black), with stability decreasing quickly for larger values of L (from [74]).

a well defined population could be studied to understand the loss mechanisms of electrons in the radiation belts. This let us understand that electrons were lost very quickly below around 1.3 Earth radii, due to atmospheric precipitation, and above around 1.8 Earth radii, because they begin to enter the slot region. Similarly, we learned that the loss mechanisms are very weak between these two bounds, which allows electrons to remain trapped for years at a time [74].

Unfortunately, while the large population of electrons injected were valuable scientifically, they had a devastating impact on satellites operating at the time. The satellites were exposed to high doses of radiation over a prolonged period of time. This caused 8 out of the 24 satellites in orbit to be damaged. Of these, 6 satellites failed completely as a result of Starfish Prime, through either solar cell degradation or damage to electrical components, mainly transistors [9,

74, 81].

Solar cell degradation led directly to the eventual failure of four satellites: TRAAC, Transit-4B, OSO-1, and Ariel-1. This made it the most dominant mechanism of satellite failure from Starfish Prime. Of these, only OSO-1 survived for more than a few months after the test, and this was because it was operated only intermittently at the time of Starfish Prime, allowing it to run on reduced power [74]. Similarly, other satellites of the time experienced significant solar cell degradation but were able to avoid failure by lowering their power usage. The main examples of this were: ANNA-1B, Injun I, and Telstar-1. Unfortunately, the last of these, alongside Explorer XV, eventually failed due to radiation damage to its transistors. Telstar-1 was a particularly unfortunate case, as it had launched only one day before the Starfish Prime test and failed completely within seven months [9].

If a similar event were to happen today, the consequences could be devastating. While modern electronic components are becoming increasingly resistant to radiation damage, they are not immune, and solar cells remain particularly vulnerable to prolonged electron exposure. Given the greatly increased number of satellites in low-Earth orbit, and the complexity of modern spacecraft's electrical components, such an event would deal severe damage to our space-based infrastructure [74, 82].

7.3 Artificial Aurora

Natural auroras are seen when charged particles, typically from the solar wind, collide with molecules in Earth's upper atmosphere. These collisions excite the atmospheric gases, which then emit light upon returning to a lower energy state. This is typically seen at high latitudes, as the Earth's magnetic field guides the solar wind towards the poles. Here, the charged particles mainly collide with oxygen to create green and red light, and with nitrogen to produce blue and purple light.

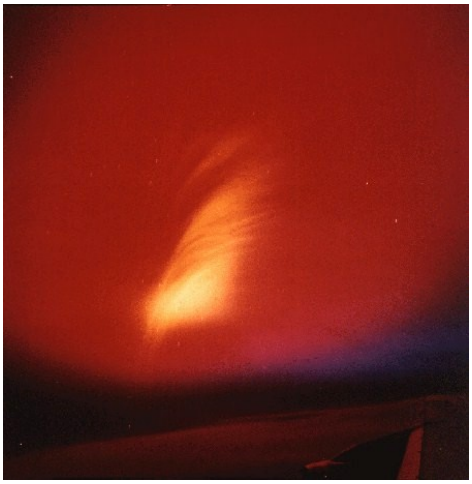


Figure 23: **Aurora seen after Starfish Prime**. Photo taken 3 minutes after the test by a nearby surveillance aircraft. Bright part shows the debris fireball while the red air glow is the aurora (from [84]).

The Starfish Prime test instead produced an artificial aurora, pictured in Figure 23. This occurred when electrons, many of which were released by the diamagnetic cavity, travelled along the Earth's magnetic field lines into the upper atmosphere, where they collided with oxygen and nitrogen molecules. Due to the long range of the geomagnetic field lines, this created auroral displays as far from the test site as New Zealand, over 6400km away. To the naked eye, a red glow covered the sky for about 12 minutes after the test, but equipment measured abnormal light levels for 3.5 hours [83].

The artificial aurora were also scientifically useful. Measurements of how the aurora's intensity changed over time allowed theoretical predictions of the lifetime of the ^{1}D state of Oxygen to be experimentally validated [83]. In this way, the atmospheric effects of Starfish Prime paved the way for major scientific discoveries, despite the destructive nature of the weapons tested.

8 Modern Implications

Dominic Starfish Prime demonstrated that outer space could be affected by terrestrial weapons, and that the effects of detonating a nuclear bomb there could pose a uniquely unpredictable and dangerous threat to mankind. The Starfish Prime test had unexpectedly widespread and long-lasting electromagnetic and magnetospheric effects [85]. The damage caused transcended national borders, contributing to the mounting political and scientific pressure seeking to regulate the testing of nuclear weapons. In the years following Starfish Prime, two major treaties were signed with the purpose of limiting the damage these tests caused: the "Partial Test Ban Treaty" and "Outer Space Treaty".

8.1 Political and Regulatory Implications

The Partial Test Ban Treaty (PTBT), formally known as the 'Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space and Under Water', was signed by the Soviet Union, UK, and the United States on 5th August 1963. The purpose of this was prohibiting nuclear weapons testing underwater, in the atmosphere, and in outer space. It was also the most significant precursor to the 'Comprehensive Nuclear Test Ban Treaty' of 1996 which finally banned the testing of nuclear weapons completely, although it has not been ratified by multiple parties including the US and Russian Federation.

Starfish Prime was certainly the most startling of the High Altitude Nuclear Explosion (HANE) tests conducted by either the US or USSR between 1958 and 1962; due to its high yield and detonation altitude, Starfish Prime indiscriminately affected large areas of the globe over long time scales. The best documented examples of this are electrical components of satellites and power grids in Hawaii being damaged by an EMP, and after effects in the magnetosphere being observed up to five years later [7–9]. Therefore, it stands to reason that Starfish Prime was the main catalyst for the inclusion of outer space in the banned testing sites under PTBT.

The second treaty signed was the Outer Space Treaty (OST), formally known as the 'Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies' [10]. The OST is regarded as the "cornerstone of international space law conventions" [86], a central point of the treaty was to ban the placement of nuclear weapons or other weapons of mass destruction in orbit, on celestial bodies, or stationed in space [10]. Starfish Prime in particular demonstrated space to be a fragile environment and our technological presence in the atmosphere to be severely affected by HANEs.

The OST recognised the potential for damage of events like Starfish Prime being used in warfare, and placed safeguards against their global-scale strategic impact. The OST reframed space as an actively regulated domain "for the benefit of all peoples" rather than a passive military backdrop to be used for strategic advantage. The OST's relevance is still felt in the 21st century, with 14200 functioning satellites in space [87] a HANE would cause disproportionate civilian harm.

8.2 Unintended Scientific Legacy

Verification of compliance with PTBT in outer space was carried out using the Vela satellite constellation, the first pair of which were launched 6 days after the treaty went into effect [88]. Each satellite was equipped with X-

ray, gamma-ray, and neutron detectors which enabled them to recognise the 'double flash' optical signature caused by the detonation of a nuclear weapon in space [89].

The double flash describes the following process which begins with a flash consisting of immediate thermal radiation occurring within microseconds of detonation. Then, the ionised shockwave which becomes opaque to visible light dims the output, this process is referred to as obscuration. Finally, the shockwave expands and cools, becoming transparent once again. The necessity for X-ray, gamma-ray, and neutron detectors is due to the obscuration phase of the double flash, which in the absence of an atmosphere, no hydrodynamic shock front forms, eliminating the optical obscuration phase entirely. This led to a need for other detection mechanisms [90].

The Vela satellite constellation never witnessed an exoatmospheric nuclear bomb detonation, rather, it was the first system to detect gamma ray bursts (GRBs) in the late 1960s [91]. Given that they have "nearly continuous and omnidirectional coverage" [92] and placement in the exoatmosphere, the Vela constellation was thus uniquely equipped for GRB detection. This potentially advanced our understanding of high-energy astrophysics by decades.

8.3 Impact on Modern Military Modelling

Starfish Prime demonstrated that an EMP is strategically viable, unpredictable, and that, at its worst, it can be seen as an electronic 'scorched earth' weapon. Hence, western strategic and military reports and models began incorporating its potential use into risk assessments of China, the USSR, and other actors. A 2008 technical report on HEMP by the Defense Technical Information Centre (DTIC) expressed concern that public and private sectors were "poorly prepared" for use of an EMP by malicious actors, and that the US more generally is "rapidly losing the technical competence in this area that it needs" [93]. Another 2020 report from DTIC expresses particular unease pertaining to China's advancement in the field of both nuclear and non-nuclear HEMP devices, writing that the Chinese military perceives nuclear EMP devices as the premier weapon for use in electronic warfare [94].

The level of technological progress necessary to carry out a nuclear triggered EMP is lower than that for other kinds of electronic and nuclear warfare, making it an attractive choice for 'rogue states' with less developed nuclear technology such as North Korea and Iran [95]. In the same paper, it is considered that due to its impact on technology rather than people, an EMP is a greater strategic or diplomatic tool for these states than deploying nuclear weapons. As discussed earlier, in 1962 and the years that followed, Starfish Prime damaged 8 out of 24 satellites in orbit [9]. A comparable event today would cripple essential services to modern, everyday life such as GPS, financial, and communications infrastructure [96].

Despite the danger an event that puts "systems on a national and international level" [97] at risk, the amount of empirical data on a HEMP and its effects has been frozen since the PTBT was signed. Due to the "uncertainty of multiple variables"[85] and lack of empirical data, models are restricted by the limited data gathered before 1963 and predict a wide range of outcomes, leading to governments and companies preparing for the determined 'worst case' scenario.

These measures are varied in critical infrastructure, such as power grids and satellites. Military satellites are radiation hardened to withstand HANEs but for commercial satellites such as the Starlink constellation this is "cost-

prohibited" [98], rendering them vulnerable to damage or failure caused by a HEMP scenario. According to a 2015 report from US Department of Energy the lack of data is a significant problem in modelling a HEMP's effect on nationwide civilian electric grids, stating: "Lack of empirical data is a known gap in research" [11]. They note that empirical research into EMP effects has prioritised the efficacy of military-grade shielding, leaving a significant data deficit regarding the resilience of unhardened civilian infrastructure, particularly the electrical grid.

The freeze in data generated is particularly impactful when considering how much electrical components have advanced over the last 60 years, with transistors in processing units shrinking from 10000nm in 1960 to sub-10nm in 2026 [99]. Moreover, electrical infrastructure has become highly interconnected and there are concerns that failure of one section of the grid will not remain isolated, but rather "induce cascading failures into other infrastructures" [60].

9 Conclusion

The development of nuclear weapon research in the 20th century was driven strongly by political tensions, highlighting the substantial impact that the Cold War had on scientific discovery and the understanding of these weapons. Starfish Prime provided new insight into the effects of high-altitude nuclear explosions that had not previously been anticipated. Unlike low-altitude explosions, where energy is dispersed primarily through thermal effects, it demonstrated that high altitude nuclear weapons could have significantly further reaching impacts through radiation belts, ionisation, and electromagnetic pulses.

The EMP observed during Starfish Prime exposed the errors in theoretical prediction of high altitude tests, jump starting the rush to develop electronic hardening and EMP resistance due to the new threat of an EMP attack. Since then, the effects of an EMP have remained a significant concern to the US Department of Defense, as shown by major investments into the prevention of an EMPs ability to disable civilian and military infrastructure.

In addition, Starfish Prime had enormous effects on the atmosphere and on space, being responsible for the formation and collapse of a diamagnetic cavity. This, along with particles from the bomb, caused the creation of an artificial radiation belt and an artificial aurora. These effects were detectable for five years after the test was carried out, with direct consequences for satellites and global communications. This exposed a huge threat to modern infrastructure but also contributed to scientific discovery through analysis of the auroras intensity.

Starfish Prime acted as the catalyst for change. As mentioned, it resulted in the Partial Test Ban Treaty and the Outer Space Treaty, preventing countries from carrying out nuclear tests in outer space. This prevented further damage to the atmosphere and to infrastructure both on the ground and in space, but it also resulted in there only being a few case studies of high altitude testing from which we can draw data. Combined with the fact that electronics have advanced dramatically since the 1960s and our growing reliance on satellites, this makes it very difficult to predict the true extent of the damage a high altitude nuclear bomb could have now.

Starfish Prime therefore acts as a cautionary example of technological advancements being tested before proper understanding. Despite the large amounts of unpredicted damage, the Starfish Prime test supplied crucial data regarding the effects of high altitude nuclear explosions and aided the development of EMP protection

mechanisms. To this day, the test remains a vital case study in nuclear physics, as well as in security policy globally.

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